AI · DIGITAL HEALTH · PREDICTIVE MONITORING

GLUCOTWIN AI

A Predictive Digital Twin for Type 2 Diabetes

Combining synthetic clinical history and dynamic health signals to explore glucose-risk prediction.

Happiest Health

Digital Twin Challenge 2026

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02 / PROBLEM STATEMENT

Healthcare Needs to Look Ahead

Diabetes monitoring often relies on measurements that have already occurred. A single reading cannot capture the complete, changing health state of a patient.

Reactive monitoring

Observes glucose measurements after changes have occurred.

Fragmented health data

Clinical history and dynamic wearable signals provide different pieces of the patient's health picture.

The opportunity

Combine both data streams to explore short-term glucose-risk prediction.

OUR CHALLENGE

Can a virtual patient help anticipate glucose risk?

OUR SOLUTION: GLUCOTWIN AI

A Predictive Digital Twin for Type 2 Diabetes

GlucoTwin AI creates a virtual representation of a patient using synthetic clinical history and dynamic health signals to explore short-term glucose-risk prediction.

HOW IT WORKS

01 — BUILD THE TWIN

Combine synthetic patient records with health-related signals.

02 — PREDICT THE RISK

Use a machine-learning model to estimate whether glucose may exceed **180 mg/dL within the next 2 hours**.

03 — SIMULATE WHAT-IF SCENARIOS

Explore how meal choices, post-meal walking, and sleep settings may affect predicted risk.

OUR GOAL

Move from simply observing glucose readings to exploring potential risks before they occur.

Proof of concept using synthetic data. Not validated for clinical diagnosis or treatment decisions.

SYSTEM ARCHITECTURE

From Synthetic Patient Data to Predictive Insights

01. DATA INPUT

Synthetic patient records
Glucose readings and health signals
Clinical indicators and lifestyle factors

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02. DATA PROCESSING

Data cleaning and preprocessing
Feature preparation
Patient-specific input selection

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03. MACHINE LEARNING ENGINE

Gradient-boosted tree model
Learns patterns from synthetic training data
Estimates short-term glucose risk

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04. DIGITAL TWIN DASHBOARD

Patient health overview
Glucose trend visualization
Predicted risk score
What-if scenario simulator

OUTPUT

An interactive proof of concept for exploring potential glucose-risk scenarios.

Data is synthetic. Model performance and predictions require independent validation before clinical use.

DATA & MACHINE LEARNING MODEL

Turning Synthetic Health Data into Predictive Insights

DATA USED

- Synthetic patient records
- Glucose measurements and health-related signals
- Clinical and lifestyle-related features

MODEL APPROACH

Gradient-Boosted Trees

A machine-learning approach used to identify patterns in the prepared data and estimate the likelihood of a future glucose-risk event.

PREDICTION TARGET

Glucose above 180 mg/dL within the next 2 hours

The model estimates whether this threshold may be crossed based on the available input features.

MODEL EVALUATION

- Reported AUROC: **0.96**
- Evaluation based on the project's synthetic-data setup
- Further testing is needed to assess generalization to real-world patient data

KEY TAKEAWAY

A promising proof of concept is a starting point—not proof of clinical accuracy.

The dataset is synthetic. The reported metric does not establish real-world clinical performance.

INTERACTIVE DASHBOARD & WHAT-IF SIMULATOR

Explore Patient Risk. Test Different Scenarios.

PATIENT HEALTH DASHBOARD

- Patient profile and clinical indicators
- Glucose trend visualization
- Predicted glucose-risk score
- Health-related information at a glance

WHAT-IF SIMULATOR

What happens when lifestyle inputs change?

Meal Settings
Explore how different meal scenarios affect the model's predicted risk.

Post-Meal Walking
Adjust walking activity to compare predicted outcomes.

Sleep Settings
Explore alternative sleep scenarios and observe changes in the model's output.

WHY IT MATTERS

- Makes model predictions easier to explore
- Supports interactive scenario comparison
- Helps demonstrate the potential of personalized digital twins

KEY MESSAGE: From a static health report to an interactive, data-driven exploration tool.

This prototype uses synthetic data. Simulated outcomes are model estimates, not guaranteed physiological effects or medical advice.

RESULTS & KEY FINDINGS

What Does Our Proof of Concept Demonstrate?

01 — MODEL PERFORMANCE

Reported AUROC: 0.96

The project reports an AUROC of 0.96 on its synthetic-data evaluation setup. This result requires further validation to establish how well the model generalizes.

02 — SHORT-TERM RISK ESTIMATION

The model estimates whether glucose may exceed **180 mg/dL within the next 2 hours**, based on the available input features.

03 — INTERACTIVE SCENARIO EXPLORATION

The dashboard allows users to adjust meal, post-meal walking, and sleep settings and compare changes in predicted risk.

04 — INTEGRATED HEALTH VIEW

The prototype brings glucose trends, patient indicators, risk estimates, and scenario simulation into one interface.

KEY FINDING

GlucoTwin AI demonstrates a way to combine machine-learning predictions with interactive digital-twin scenario exploration.

IMPORTANT LIMITATION

The data is synthetic. The reported AUROC is not evidence of clinical effectiveness, and simulated changes do not prove real physiological outcomes.

LIMITATIONS, ETHICS & RESPONSIBLE AI

Innovation Must Be Matched by Responsibility

DATA LIMITATIONS

The prototype uses synthetic patient data, which may not represent the diversity and complexity of real-world patients.

MODEL LIMITATIONS

The reported AUROC of 0.96 comes from the project's synthetic-data evaluation. It does not establish clinical accuracy or generalization to real patients.

SIMULATION LIMITATIONS

Changes in predicted risk after adjusting meal, walking, or sleep settings are model outputs—not proof of actual physiological effects.

PRIVACY & SECURITY

Real-world deployment would require appropriate protection of sensitive health information, secure data handling, and responsible access controls.

HUMAN OVERSIGHT

GlucoTwin AI is an experimental decision-support concept, not a replacement for healthcare professionals or medical judgment.

OUR RESPONSIBLE AI PRINCIPLE

Transparent predictions. Clear limitations. Human-centered healthcare.

GlucoTwin AI is a proof of concept and is not intended for diagnosis, treatment, or independent clinical decision-making.

FUTURE SCOPE & POTENTIAL IMPACT

From a Proof of Concept to a More Personalized Healthcare Future

01 — REAL-WORLD VALIDATION

Evaluate the model using representative, ethically sourced clinical data and independent testing before considering clinical use.

02 — PERSONALIZED DIGITAL TWINS

Explore how individual health histories and continuously updated signals could support more patient-specific risk estimates.

03 — SMARTER PREDICTION

Investigate additional features, improved calibration, and robust evaluation to make risk estimates more reliable.

04 — CONNECTED HEALTH SYSTEMS

Explore integration with wearable devices, electronic health records, and clinician-facing workflows—with appropriate privacy safeguards.

05 — CLINICIAN-IN-THE-LOOP DESIGN

Study how transparent predictions and human oversight could support healthcare professionals without replacing their judgment.

OUR VISION

Build a future where health data helps us anticipate potential risks—not just observe them.

These are future development opportunities, not capabilities or clinical benefits already demonstrated by the current prototype.

GLUCOTWIN AI

ANTICIPATE. EXPLORE. PERSONALIZE.

A Predictive Digital Twin for Type 2 Diabetes

GlucoTwin AI combines synthetic health data, machine-learning-based glucose-risk estimation, and interactive what-if simulation in one dashboard.

WHAT WE BUILT

- A gradient-boosted tree prediction model
- A patient-focused health dashboard
- A meal, walking, and sleep scenario simulator
- A proof of concept for exploring short-term glucose risk

OUR TAKEAWAY

The future of digital health is not just about collecting data. It's about turning data into meaningful, responsible insights.

THANK YOU!